Vasamsetti John babu

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.65
NARYANA JUNIOR COLLEGE	Class XII (BOARD OF INTERMEDIATE EDUCATION ANDHRA PRADESH)	2018	95.90%
OXFORD TALENT SCHOOL	Class X (BOARD OF SECONDARY EDUCATION)	2016	9.80%

WORK EXPERIENCE & INTERNSHIP

Assistant Manager - Projects (P&C) | Tata Steel Ltd

- **Executed project activities** for a **350 TPH reheating furnace**, interpreting P&IDs and engineering drawings to plan, schedule and track **piping and equipment installation** progress across the project site.
- **Managed material procurement and site execution**, including **MRQ preparation, vendor coordination, material tracking**, and installation of **pumps, fans, valves and process piping**.
- **Coordinated construction, testing and commissioning** through process-piping **pressure-testing, equipment trials and pre-commissioning**, working closely with **cross-functional teams and contractors**.
- **Controlled site operations and project documentation**, covering **work permits, safety practices,** milestone tracking, inspection records, as-built drawings and **commissioning dossiers**.

Industrial Internship -- Nagarjuna Sagar Hydro-Electric Scheme, TSGENCO

- Studied the operation of an **815.6 MW hydroelectric power plant** comprising **8 units**, **Francis turbine**, generator, penstock, draft tube, governor and auxiliary systems, and understood operating modes including **turbine and reverse-pump modes**.
- Gained **plant-level exposure** through facility visits, observing mechanical maintenance practices and the importance of **water sealing, lubrication systems and safety procedures** for reliable hydro-turbine operation.

PROJECTS

Experimental Study and Bayesian Optimization of Dual-Fuel Combustion in a CI Engine

- **Conducting experimental studies** on dual-fuel combustion in a CI engine, evaluating **combustion characteristics, engine performance and emissions** under varying operating conditions.
- **Developing an ML-based predictive model** to establish relationships between **engine operating parameters, performance and emissions** using experimental data.
- **Applying Bayesian optimization** to identify optimal engine operating conditions for improved **performance and emissions characteristics**.

Compressible Flow Simulations

- **Designed a supersonic converging-diverging nozzle** using the **Method of Characteristics (MOC)**, followed by **CFD simulation** using **commercial software (ANSYS Fluent)** to examine Mach number and pressure distributions.

Operational Mode Classification & Fuel Efficiency Forecasting

- Performed **feature engineering** and **model evaluation** using **Gradient Boosting** and **Random Forest** to classify engine **operating modes** and **predict fuel efficiency** from operating parameters.

Weight Reduction of Electric Bike Chassis

- **Performed topology optimization of an electric-bike chassis using ANSYS** to reduce structural weight while maintaining **structural integrity and load-carrying capability**.
- **Evaluated the optimized chassis through structural analysis and simulation**, comparing its **deformation and stress response** to assess its suitability for lightweight design.

COURSES & CERTIFICATIONS

Courses: Incompressible Fluid Flow · Compressible Fluid Flow · Advanced Heat Transfer · Computational Fluid Dynamics (CFD) · Machine Learning · Spray & Atomization · Internal Combustion Engines

Introduction to Programming with MATLAB-Vanderbilt University (Coursera)

Python for Mechanical Engineers -- Skill-Lync

Energy Literacy Training -- Energy Swaraj Foundation

Himalayan Outdoor Leadership Program -- Tata Steel Adventure Foundation (TSAF)

- **Completed a 10-day Outdoor Leadership Program** conducted by **Tata Steel Adventure Foundation (TSAF)**, including **Mugdha Top (11,400 ft)** trek and structured team-based outdoor leadership activities.
- **Trained in rappelling, caving, rock climbing, water sports and navigation**, with emphasis on **teamwork, communication, adaptability and resilience**.

SKILLS

- **Engineering & Computational Tools: MATLAB, Python, ANSYS/CFD, AutoCAD, SolidWorks**; engineering analysis, modelling, simulation and data analysis.
- **Professional Skills: Project Management, Problem Solving, Technical Communication, Team Collaboration, Adaptability**.

POSITION OF RESPONSIBILITY

Teaching Assistant | IIT Hyderabad

- Supported teaching and assessment of **120+ students** in **Mechanics of Solids** and **Fluid Mechanics Laboratory** as part of a **5-member TA team**, handling invigilation, evaluation, re-verification and mentoring students during laboratory experiments.

Online Engineering Tutor & SME | Chegg & Course Hero

- Applied technical problem solving to deliver verified, high-accuracy solutions for hundreds of academic queries across online tutoring websites to support students.

HOBBIES & EXTRACURRICULAR

- Cricket, Volleyball, Travelling.
- Represented JNTUHCEM College cricket team; **JNTUH B-Zone Champions (2019-20)** and **Runner-up in the Eenadu Cricket Tournament**.
- Participated in **10km running events in Jamshedpur and Kolkata**.
- Participated in trekking expeditions in the **Himalayas and Dalma Hills**, gaining experience in challenging terrain, endurance and outdoor teamwork.